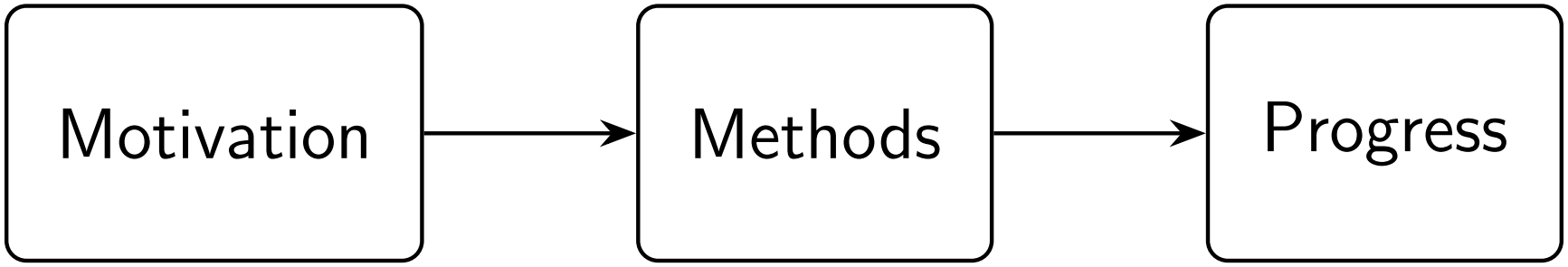
# A Search for CPT and Lorentz Invariance Violations Using Neutrinos at the NO$\nu$A Near Detector

## Introduction and Overview

Romila Gargeshwari, Krishna Lakkaraju, Brian Rebel

NOvA Collaboration Meeting
2 June 2025

# Overview



# Lorentz Invariance and CPT Symmetry

- **Lorentz Invariance:** Laws of physics are the same across all inertial frames of reference.
- **CPT symmetry**: Invariance under charge conjugation (C), parity inversion (P), *and* time reversal (T).
- Violations could imply new physics beyond the Standard Model.

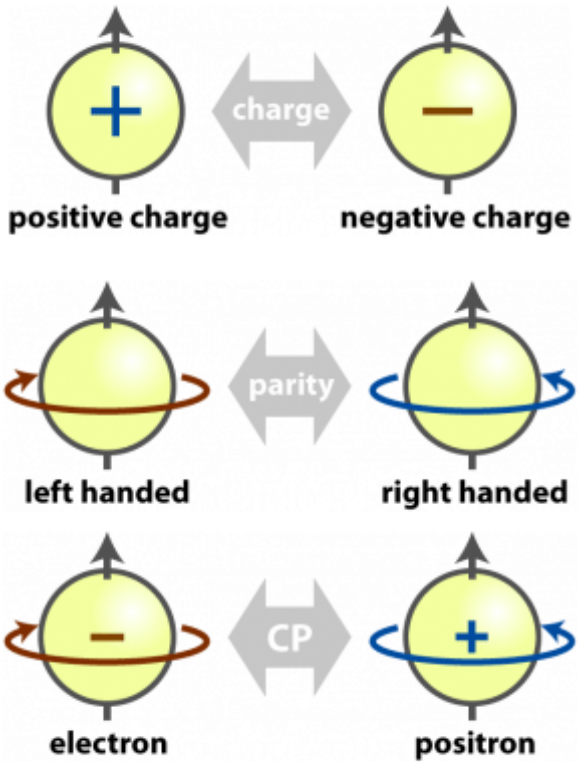



*Image adapted from briankoberlein.com*

# The Standard Model Extension (SME)

**What is the SME?**

- Extends Standard Model by including all possible terms that violate Lorentz and CPT symmetry
- These terms arise in some quantum gravity theories at the Planck scale ($\sim 10^{19}$ GeV).

**Effective Hamiltonian***

$$\mathcal{H}_{\text{eff}} = \frac{1}{E}\left[(a_L)_\mu p^\mu - (c_L)_{\mu\nu} p^\mu p^\nu\right]$$

- $E$ is the true neutrino energy
- $p^\mu$ is the four-momentum of the neutrino
- $(a_L)$ and $(c_L)$ contain energy and direction dependent, LV violating coefficients
  - $(a_L)_\mu$ is a four component, rank-one tensor
  - $(c_L)_{\mu\nu}$ is a sixteen component, rank-two tensor

---

* *Kostelecký and Mewes, Phys. Rev. D* **69**, *016005 (2004), page 2.*

# Why NO$\nu$A?

- Mass-induced oscillations are suppressed at the ND baseline.
- NO$\nu$A $\nu_e$ sample size may provide improved sensitivity to those SME coefficients
- Able to look for low probability $\nu_\mu \to \nu_e$ transitions due to the beam's flavor composition.
- This study is **the first attempt** at a search for Lorentz-Invariance violations at NO$\nu$A.

| $\nu_\mu \to \nu_\mu$ | $\nu_\mu \to \nu_e$ | $\nu_e \to \nu_e$ |
|---|---|---|
| MINOS | MiniBooNE | MiniBooNE |
| Previously Studied[1] | Channels for Possible Improvement[2] | |

[1] *Adamson et al., Phys. Rev. Lett.* **105**, *151601 (2010)*
[2] *A. A. Aguilar-Arevalo et al., Phys. Lett. B* **718**, *1303–1308 (2013)*

# Sidereal Time

- A **solar day** is the time it takes for the Sun to return to the same position in the sky
- A **sidereal day** is the length of one rotation about the Earth's axis
- The sidereal day is approximately four minutes shorter than the solar day

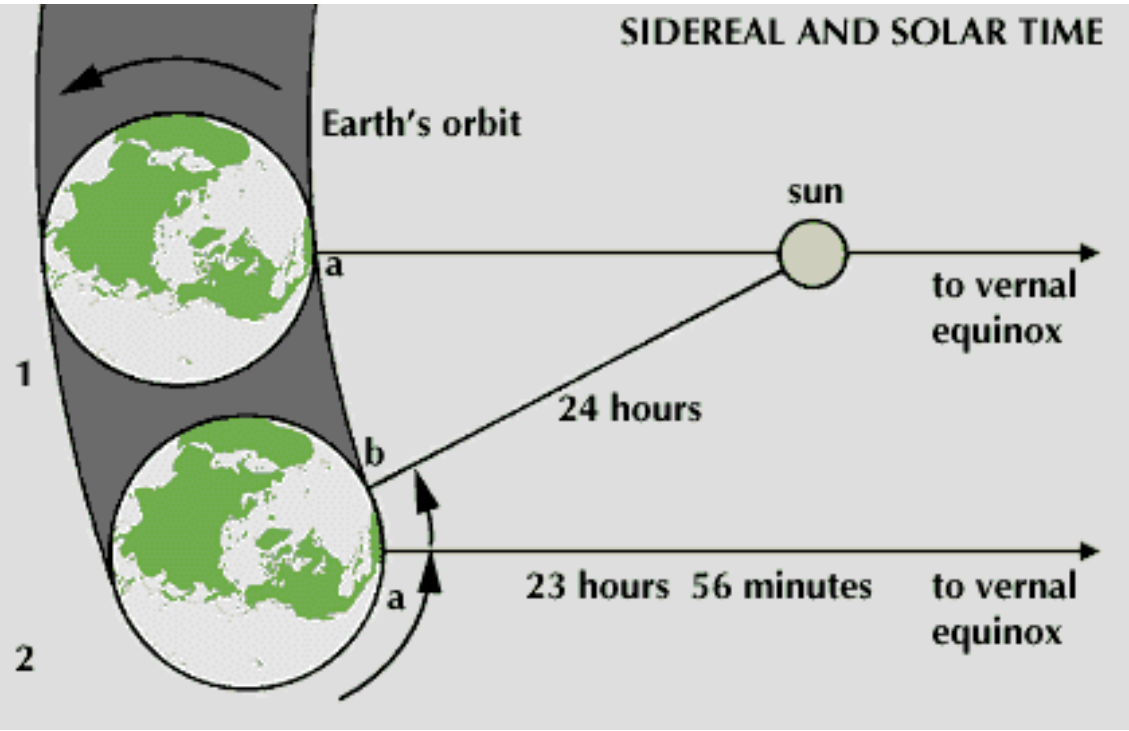



*Image adapted from www.schoolphysics.co.uk (Solar and Sidereal Days).*

# Sidereal Modulation

- If LV exists, we will observe these effects as sidereal modulations on the neutrino event rate
- Rotation of the NuMI beam in the background field allows access to sidereal modulation
- The directional cosines describe the orientation of the NuMI beam in the background:

$$\begin{pmatrix} \hat{N}^X \\ \hat{N}^Y \\ \hat{N}^Z \end{pmatrix} = \begin{pmatrix} \cos\chi \sin\theta \cos\phi + \sin\chi \cos\theta \\ \sin\theta \sin\phi \\ -\sin\chi \sin\theta \cos\phi + \cos\chi \cos\theta \end{pmatrix}$$

Colatitude Azimuth Zenith

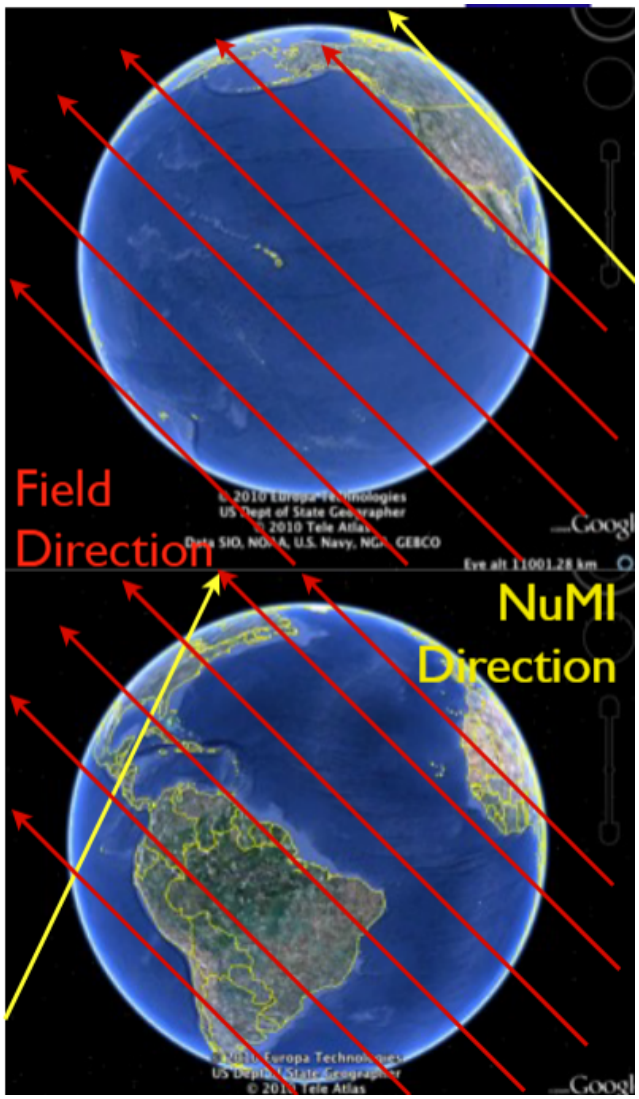



*Image courtesy of Prof. Brian Rebel, UW–Madison $\nu$-group, 2011.*

# SME Hamiltonian Expansion

- $\mathcal{H}_{\text{eff}}$ expansion in terms of sidereal time to first order*

$$\mathcal{H}_{\text{eff}} = A_s \sin(\omega_\oplus T_\oplus) + A_c \cos(\omega_\oplus T_\oplus) + B_s \sin(2\omega_\oplus T_\oplus) + B_c \cos(2\omega_\oplus T_\oplus)$$

- $A_s$, $A_c$, $B_s$, and $B_c$ are dependent on the $\hat{N}^X$, $\hat{N}^Y$, $\hat{N}^Z$, and the relevant SME components

$$(a_L)^X,\ (a_L)^Y,\ (c_L)^{TX},$$
$$(c_L)^{TY},\ (c_L)^{XX},\ (c_L)^{XY},$$
$$(c_L)^{XZ},\ (c_L)^{YY},\ (c_L)^{YZ}$$

- We wish to constrain these coefficients by checking for event rate modulation in the first four sidereal harmonics.

---

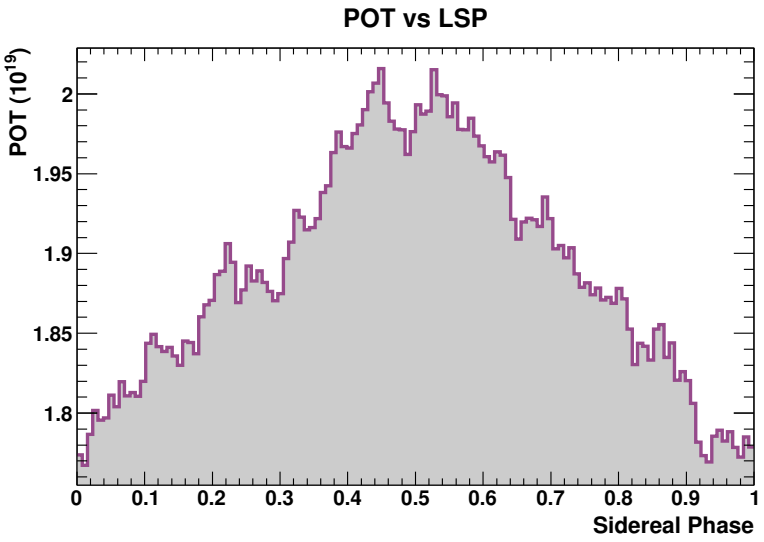
* *Kostelecký and Mewes, Phys. Rev. D* **69**, *016005 (2004), page 2.*

# Getting Sidereal Time

- Collate each ND spill's UNIX time and POT into TTree from*:

```
prod_sumdecaf_R20-11-25-prod5.1reco.a_nd_genie_N1810j0211a_nonswap_fhc_nova_v08_full_v1_numu2020
prod_sumdecaf_development_nd_genie_N1810j0211a_nonswap_fhc_nova_v08_full_v1_nue2020prod5.1
```

- UNIX time is converted to local sidereal time with `AstroPy`
  - Time conversion accepts a UNIX value and longitude to account for time zones
  - LSP = Local Sidereal Time / 24 Sidereal Hours
- POT vs LSP shows variation in beam intensity with the sidereal day

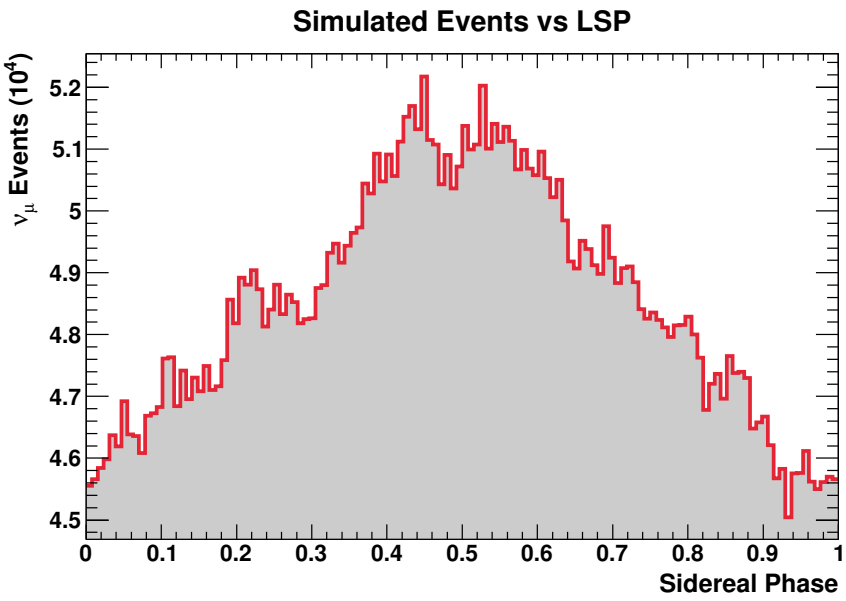


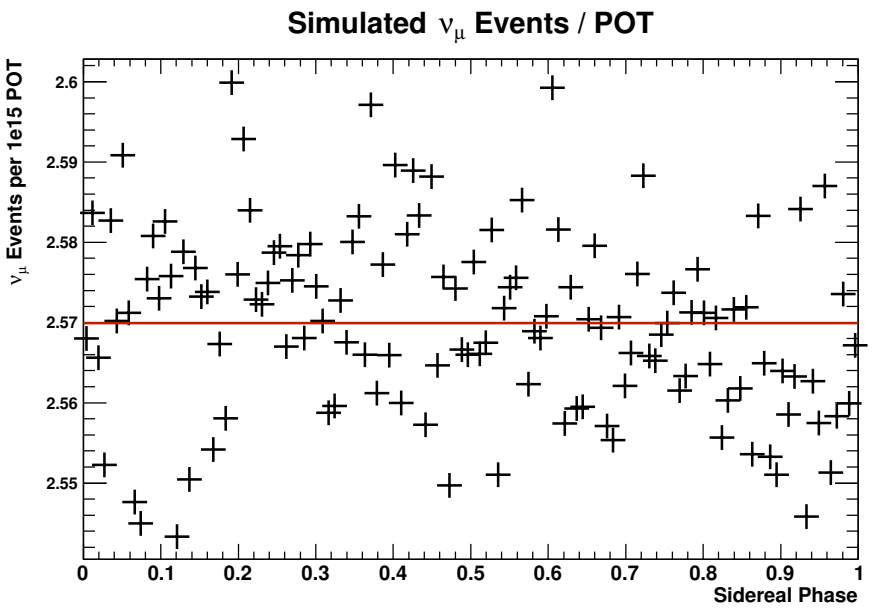

*SAM Definitions courtesy of Anna Cooleybeck, UW-Madison $\nu$-group, 2025.*

# Event Simulation

- Events = Event Rate × Total POT
- Event Rate* = $2826066\ \nu_\mu$ Selections $/11 \times 10^{20}$ POT
- Assign each event a random phase from POT vs LSP

- Divide Events vs LSP by POT vs LSP
- Normalization removes the effect of variation in beam intensity.

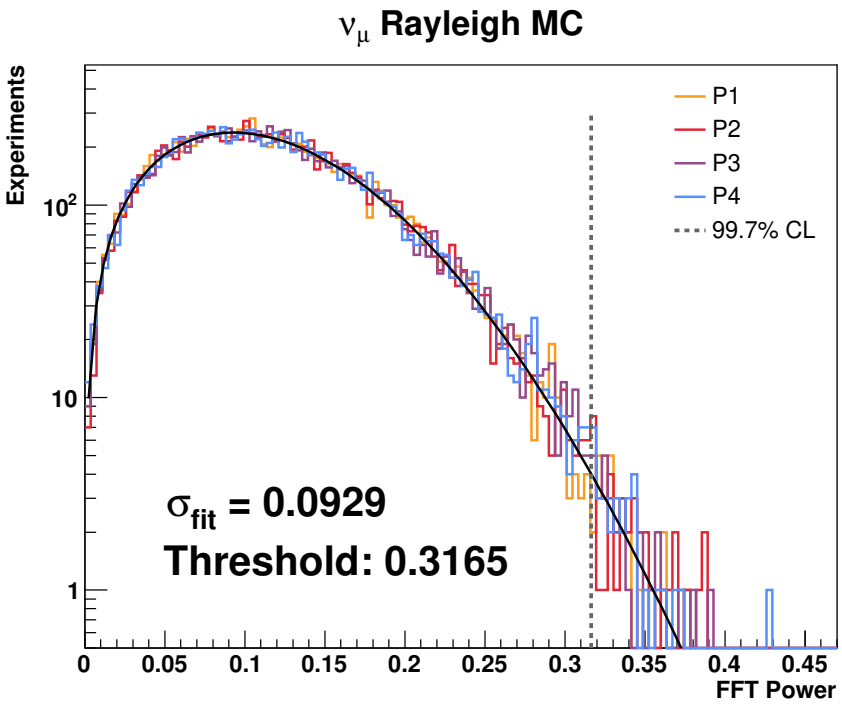



*NOνA Collaboration, Phys. Rev. Lett. **134**, 081804 (2025), page 5.*

# Rayleigh Distribution

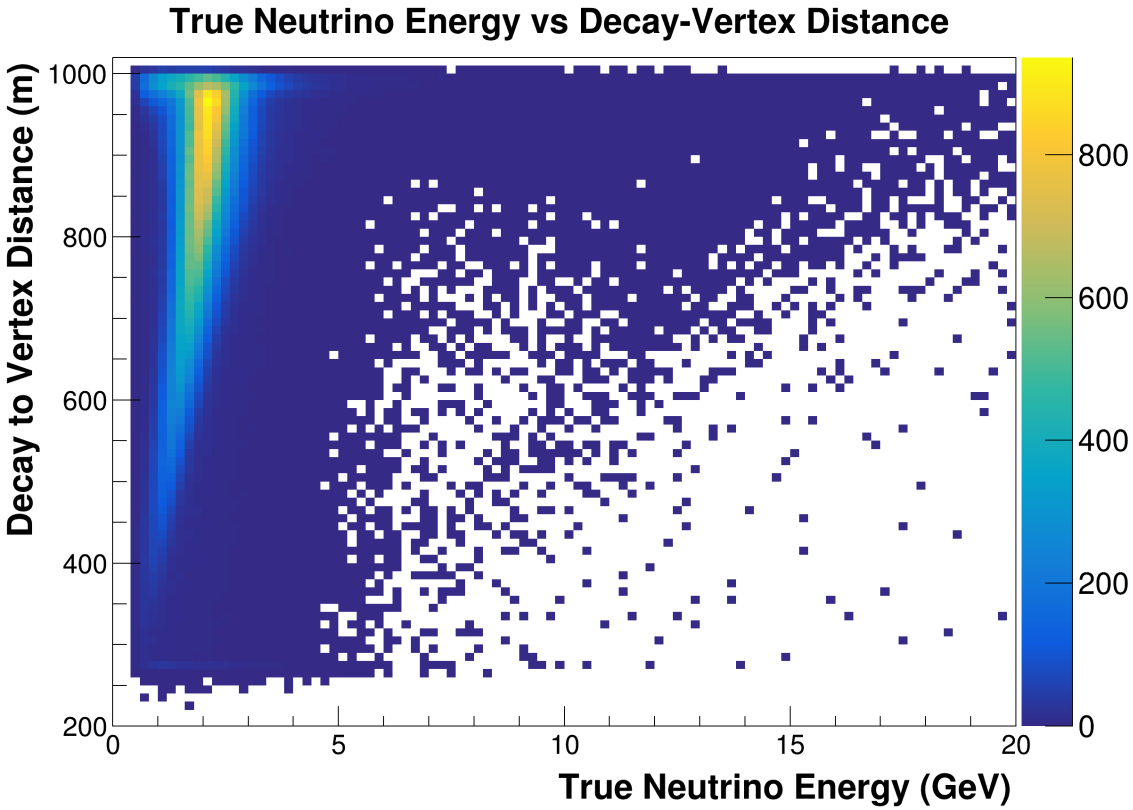
- In each of $10^4$ experiments, we fill a new events histogram with randomized LSP, normalized by POT vs LSP
- Randomizing LSP values removes any possible sidereal modulation
- FFT power computed on normalized histogram
- Shows the four harmonic power values in each simulated experiment
- The Rayleigh fit lets us understand the spectrum of possible FFT amplitudes




$$f(x) = \frac{x}{\sigma^2}\exp\left(-\frac{x^2}{2\sigma^2}\right), \quad x > 0$$

# Energy-Baseline Distribution

- Distribution of events vs true energy (GeV) and decay-vertex distance (m)
- Variation in $L$ is significant at the ND due to short baseline
- Used to randomly sample $E$ and $L$ for LV-weighted simulated events



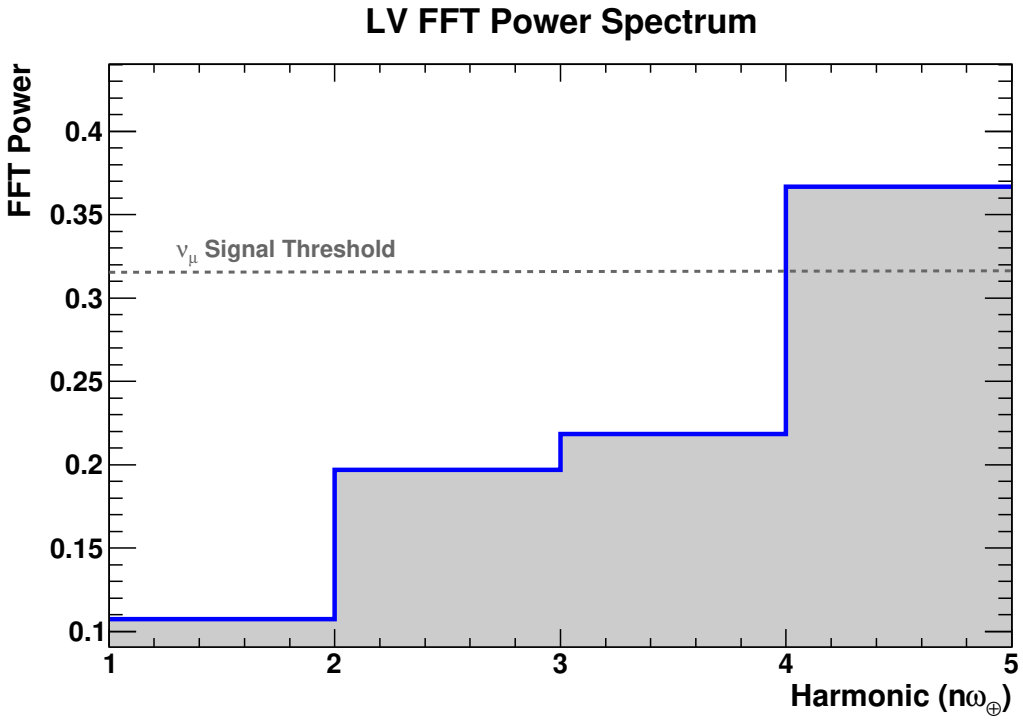

* *Histogram courtesy of Anna Cooleybeck, UW-Madison $\nu$-group, 2025.*

# Establishing SME Limits

- Set one SME coefficient non-zero to inject LV signal
- For each event, compute LV-driven oscillation weight

$$P_{\mathrm{LV}}(E, L, \omega_{\oplus} T_{\oplus}) = 1 - \frac{L^2}{(\hbar c)^2} |\mathcal{H}_{\mathrm{eff}}|^2$$

- Fill weights into a histogram, then divide by POT vs LSP and compute FFT
- Determine the smallest SME value at which FFT signal $>$ threshold

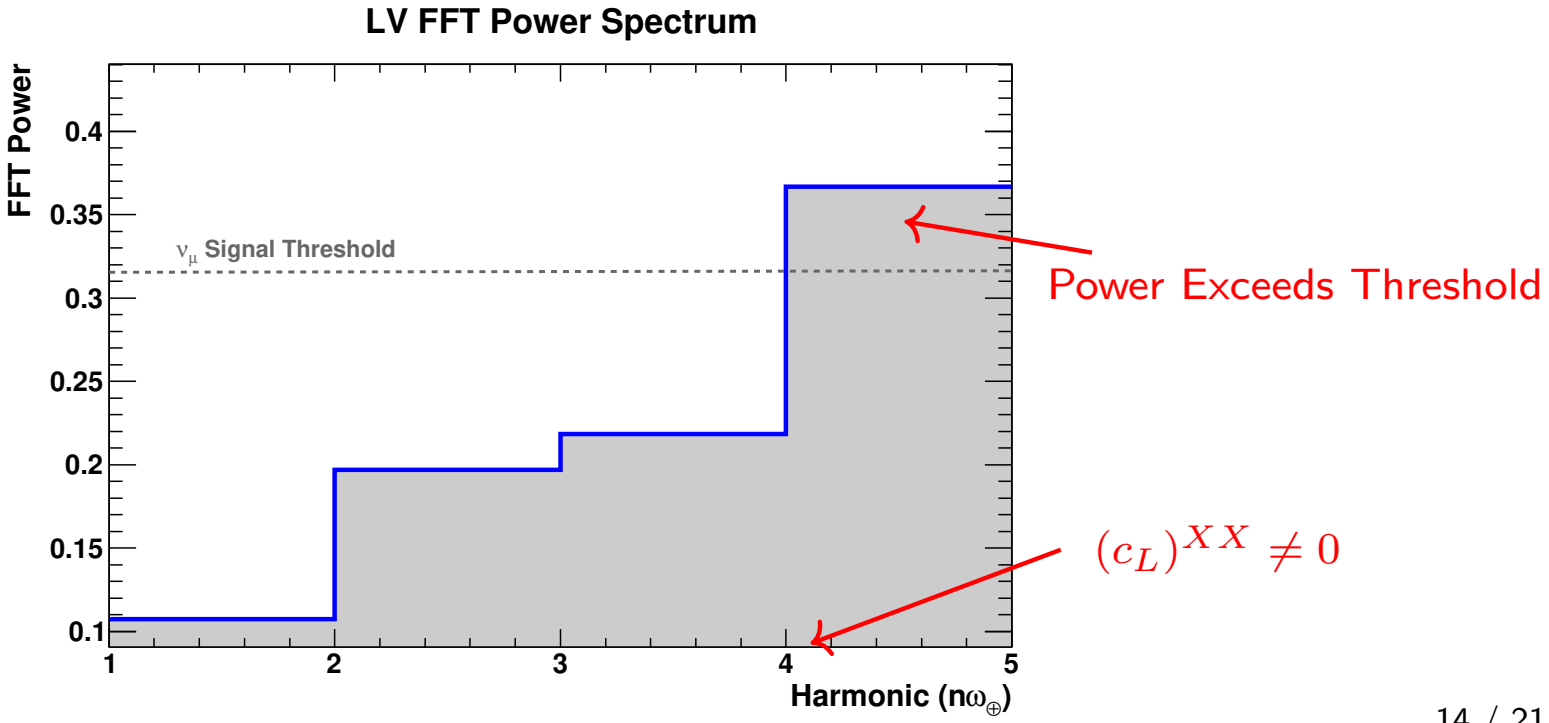


# Establishing SME Limits

- Set one SME coefficient non-zero to inject LV signal
- For each event, compute LV-driven oscillation weight

$$P_{\mathrm{LV}}(E, L, \omega_{\oplus} T_{\oplus}) = 1 - \frac{L^2}{(\hbar c)^2} |\mathcal{H}_{\mathrm{eff}}|^2$$

- Fill weights into a histogram, then divide by POT vs LSP and compute FFT
- Determine the smallest SME value at which FFT signal > threshold



# Next Steps

**Current Progress**

- Tools established: `AstroPy`, `NumPy`, `SciPy`, `pyROOT`
- Integrated Timing and POT info
- SME weight calculation determined
- Developed SME limit constraint calculation

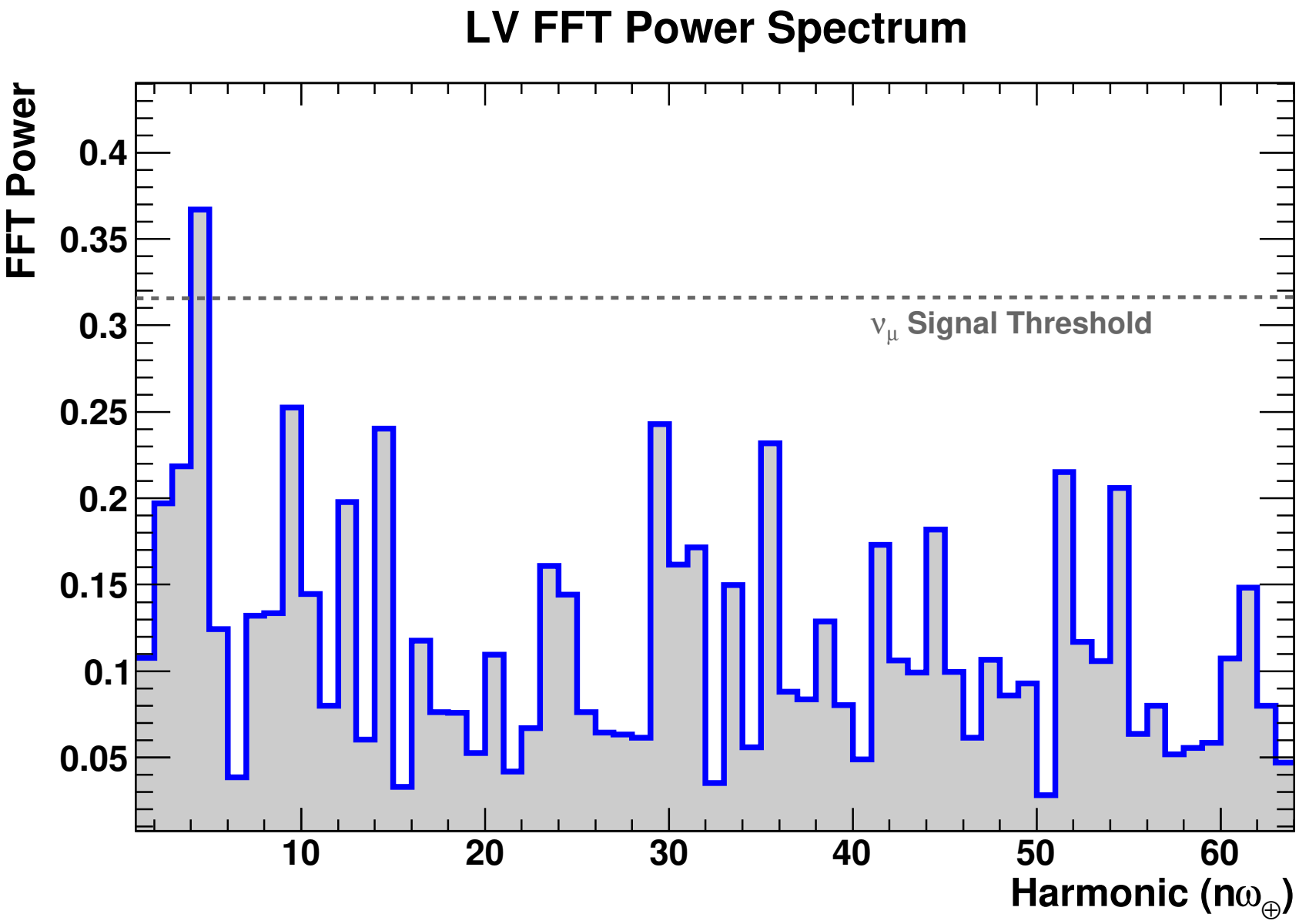
**Next Steps**

- Include the $\nu_e$ channel in to the analysis
- Finalize the analysis framework

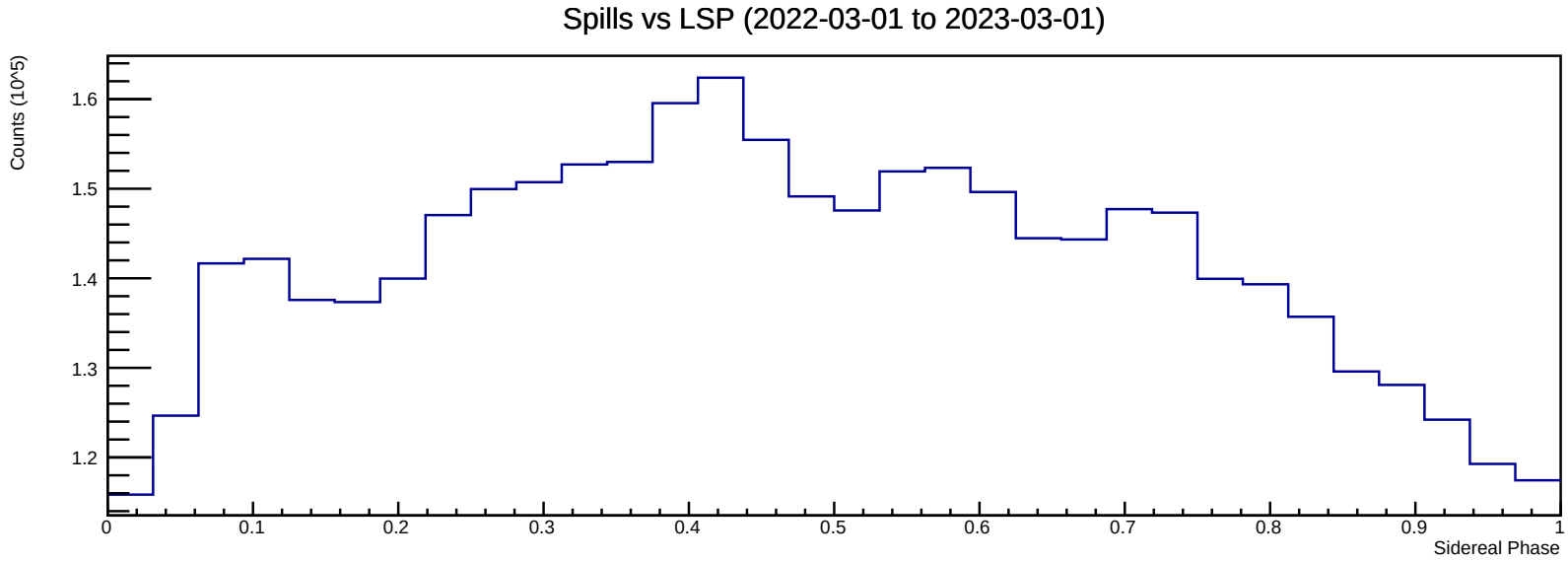
# Thank you for listening!

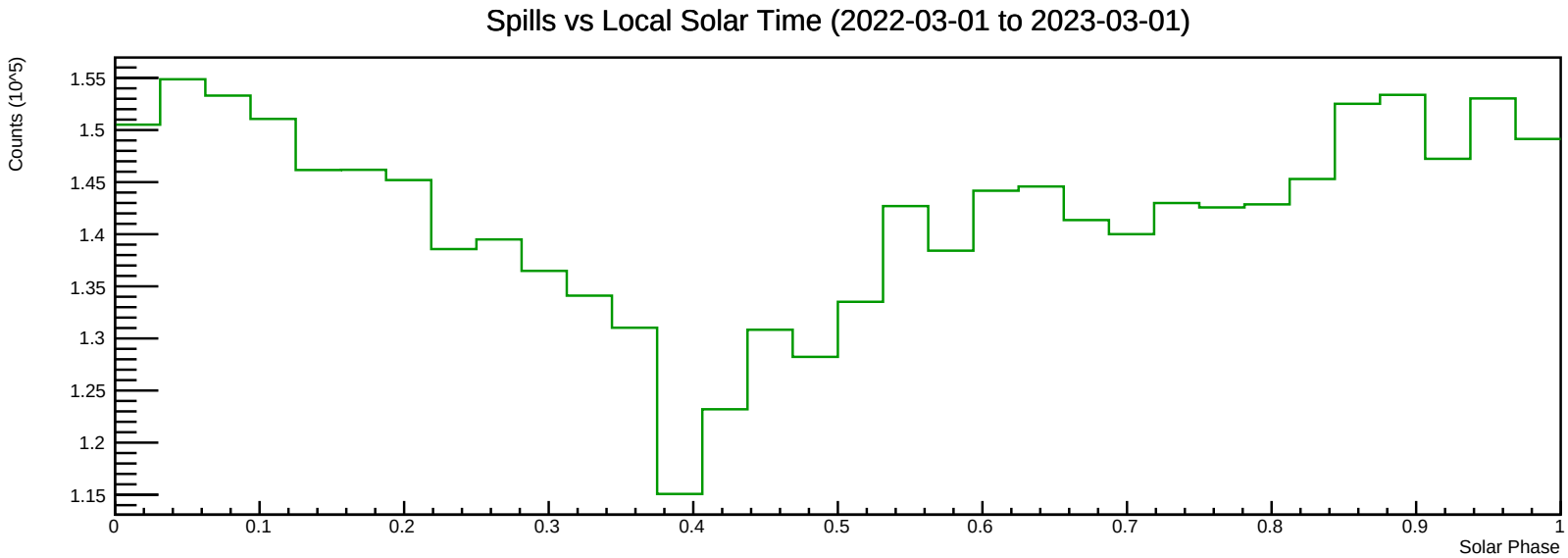
We gratefully acknowledge the many contributions from the members of the UW-Madison $\nu$-group, NO$\nu$A Collaboration, and FermiLab staff.

# Full SME FFT Plot

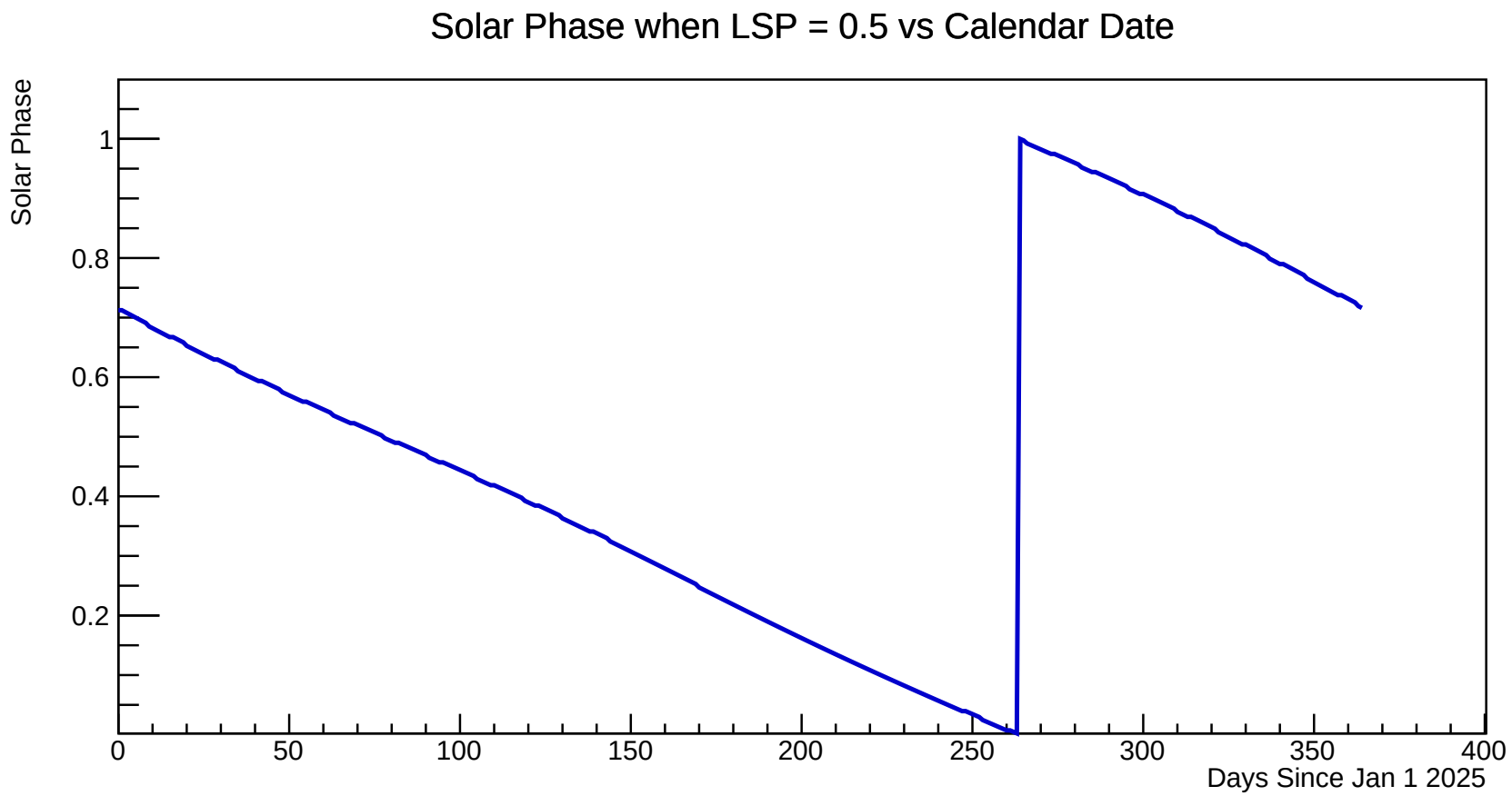


# Sidereal vs. Solar Time



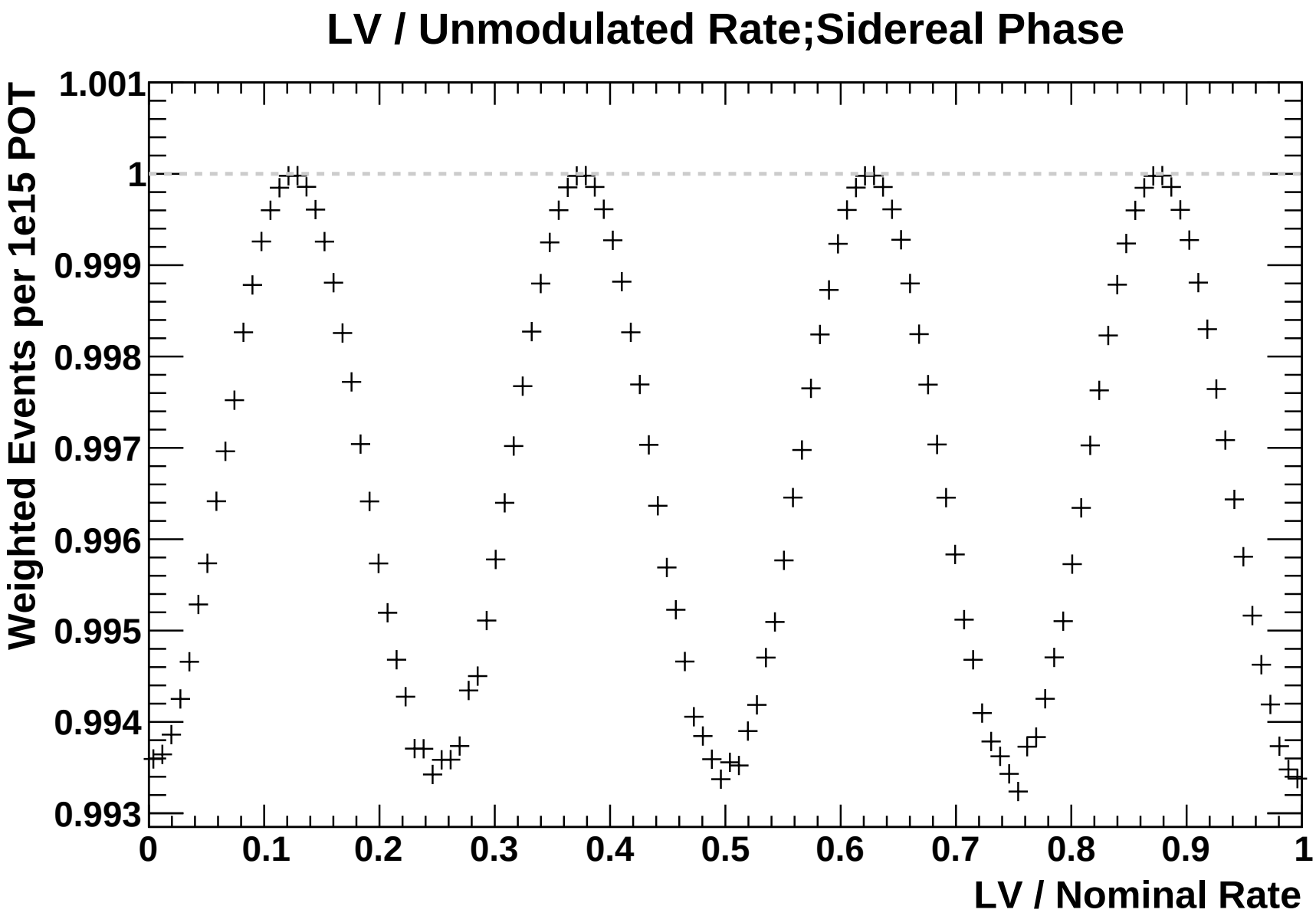
# Sidereal vs. Solar Time Cont.



# SME Intermediate Calculations

$$
\begin{aligned}
(\mathcal{A}_s^{(0)})_{ab} &= \hat{N}^Y (a_L)^X_{ab} - \hat{N}^X (a_L)^Y_{ab}, \\
(\mathcal{A}_s^{(1)})_{ab} &= -2\hat{N}^Y (c_L)^{TX}_{ab} + 2\hat{N}^X (c_L)^{TY}_{ab} \\
&\quad + 2\hat{N}^Y \hat{N}^Z (c_L)^{XZ}_{ab} - 2\hat{N}^X \hat{N}^Z (c_L)^{YZ}_{ab}, \\
(\mathcal{A}_c^{(0)})_{ab} &= -\hat{N}^X (a_L)^X_{ab} - \hat{N}^Y (a_L)^Y_{ab}, \\
(\mathcal{A}_c^{(1)})_{ab} &= 2\hat{N}^X (c_L)^{TX}_{ab} + 2\hat{N}^Y (c_L)^{TY}_{ab} \\
&\quad - 2\hat{N}^X \hat{N}^Z (c_L)^{XZ}_{ab} - 2\hat{N}^Y \hat{N}^Z (c_L)^{YZ}_{ab},
\end{aligned}
$$

$$
\begin{aligned}
(\mathcal{B}_s^{(1)})_{ab} &= \hat{N}^X \hat{N}^Y \big((c_L)^{XX}_{ab} - (c_L)^{YY}_{ab}\big) \\
&\quad - \big(\hat{N}^X \hat{N}^X - \hat{N}^Y \hat{N}^Y\big)(c_L)^{XY}_{ab}, \\
(\mathcal{B}_c^{(1)})_{ab} &= -\tfrac{1}{2}\big(\hat{N}^X \hat{N}^X - \hat{N}^Y \hat{N}^Y\big)\big((c_L)^{XX}_{ab} - (c_L)^{YY}_{ab}\big) \\
&\quad - 2\hat{N}^X \hat{N}^Y (c_L)^{XY}_{ab}.
\end{aligned}
$$

$$
\begin{aligned}
(\mathcal{C})_{ab} &= (\mathcal{C}^{(0)})_{ab} + E(\mathcal{C}^{(1)})_{ab}, \\
(\mathcal{A}_s)_{ab} &= (\mathcal{A}_s^{(0)})_{ab} + E(\mathcal{A}_s^{(1)})_{ab}, \\
(\mathcal{A}_c)_{ab} &= (\mathcal{A}_c^{(0)})_{ab} + E(\mathcal{A}_c^{(1)})_{ab}, \\
(\mathcal{B}_s)_{ab} &= E(\mathcal{B}_s^{(1)})_{ab}, \quad (\mathcal{B}_c)_{ab} = E(\mathcal{B}_c^{(1)})_{ab}
\end{aligned}
$$

# LV / Unmodulated Ratio